

Real-Time Portfolio VaR

Monitoring System:

Balancing Computational Speed and Risk Accuracy

IE 420 - Industrial Engineering Systems Analysis

Final Project Report

[Your Name Here]

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Abstract

This report presents the design and implementation of a near real-time portfolio Value-at-Risk (VaR) monitoring system applied to a five-asset equity portfolio consisting of AAPL, MSFT, NVDA, SPY, and QQQ. The system updates VaR estimates every minute using historical 1-minute intraday price data and compares two parametric estimators—rolling sample covariance and Exponentially Weighted Moving Average (EWMA) covariance—as well as a Monte Carlo simulation approach using Geometric Brownian Motion (GBM). The central engineering question is: how does computational cost scale with model complexity, and at what accuracy cost does a faster model come? Results are presented for both 95% and 99% confidence levels, with backtesting conducted via the Kupiec Proportion of Failures (PoF) test. A Plotly/Dash interactive dashboard provides a real-time view of portfolio value, VaR estimates, exceedances, and latency.

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1 Introduction and Motivation

Value-at-Risk is the most widely used quantitative risk measure in the financial industry. At a high level, VaR_α answers the question: “What is the maximum dollar loss I should expect over a 1-day horizon with probability α ?” Regulators such as the Basel Committee on Banking Supervision mandate VaR calculation for internal capital adequacy models ([Basel Committee on Banking Supervision, 1996](#)).

Traditional VaR systems are computed end-of-day on close prices. However, intraday price movements can cause substantial portfolio risk fluctuations; a portfolio that is within VaR at market open may violate risk limits by midday. This motivates the need for near real-time VaR monitoring.

The key engineering challenge is the trade-off between:

1. **Computational speed** — VaR must be recalculated every minute or faster, imposing hard latency constraints.
2. **Risk accuracy** — simpler (faster) models make stronger distributional assumptions that may underestimate tail risk.

This project implements and benchmarks three VaR approaches at increasing computational cost: (1) Parametric VaR with rolling sample covariance, (2) Parametric VaR with EWMA covariance, and (3) Monte Carlo VaR via GBM.

2 Technical Background

2.1 Value-at-Risk

For a portfolio with value V_p and a return distribution F , the VaR at confidence level $\alpha \in (0, 1)$ is:

$$\text{VaR}_\alpha = \inf\{\ell \in \mathbb{R} : P(L > \ell) \leq 1 - \alpha\}, \quad (1)$$

where $L = -\Delta V_p$ is the random loss over the holding period ([McNeil et al., 2015](#)). In this project we use a 1-day holding period and report VaR in dollar terms.

2.2 Parametric (Variance-Covariance) VaR

Under the assumption that portfolio log returns are normally distributed, VaR simplifies to a closed-form expression involving only the portfolio mean and standard deviation. For a zero-mean log-return assumption:

$$\text{VaR}_{\alpha,t} = z_{\alpha} \cdot \sigma_{p,t} \cdot V_{p,t}, \quad (2)$$

where $z_{\alpha} = \Phi^{-1}(\alpha)$ is the standard normal quantile (e.g. $z_{0.99} \approx 2.326$), $\sigma_{p,t}$ is the portfolio volatility, and $V_{p,t}$ is the portfolio value at time t (Jorion, 2007).

Portfolio volatility

Given fixed share counts $\{n_i\}$ and current prices $\{P_{i,t}\}$:

$$V_{p,t} = \sum_{i=1}^N n_i P_{i,t}, \quad (3)$$

$$w_{i,t} = \frac{n_i P_{i,t}}{V_{p,t}}, \quad (4)$$

$$\sigma_{p,t} = \sqrt{\mathbf{w}_t^{\top} \Sigma \mathbf{w}_t}, \quad (5)$$

where Σ is the $N \times N$ covariance matrix of daily log returns and $\mathbf{w}_t = (w_{1,t}, \dots, w_{N,t})^{\top}$.

2.3 Covariance Estimation

2.3.1 Rolling Sample Covariance

The simplest estimator uses the sample covariance of the most recent W trading days of log returns:

$$\hat{\Sigma}_t^{\text{roll}} = \frac{1}{W-1} \sum_{s=t-W}^{t-1} \mathbf{r}_s \mathbf{r}_s^{\top}, \quad (6)$$

where $\mathbf{r}_s$ is the demeaned return vector on day s and $W = 252$. All observations in the window receive equal weight.

2.3.2 EWMA Covariance (RiskMetrics)

The Exponentially Weighted Moving Average (EWMA) estimator assigns exponentially decaying weights to past observations ([J.P. Morgan/Reuters, 1996](#)):

$$\hat{\Sigma}_t = \lambda \hat{\Sigma}_{t-1} + (1 - \lambda) \mathbf{r}_{t-1} \mathbf{r}_{t-1}^\top, \quad (7)$$

where $\lambda = 0.94$ is the RiskMetrics daily decay factor. This estimator responds more quickly to market regime changes than the rolling estimator.

2.4 Monte Carlo VaR via Geometric Brownian Motion

Under the risk-neutral GBM assumption, asset prices over one trading day evolve as:

$$\frac{\Delta P_i}{P_i} \approx \mu_i \Delta t + \sigma_i \sqrt{\Delta t} \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1), \quad (8)$$

with correlated shocks $\varepsilon \sim \mathcal{N}(0, \Sigma)$. Correlated shocks are generated via the Cholesky decomposition $\Sigma = \mathbf{L}\mathbf{L}^\top$:

$$\varepsilon = \mathbf{L}z, \quad z \sim \mathcal{N}(0, \mathbf{I}). \quad (9)$$

For M simulated paths, the Monte Carlo VaR at confidence α is the $(1 - \alpha)$ -quantile of the empirical loss distribution:

$$\widehat{\text{VaR}}_\alpha^{\text{MC}} = -\text{quantile}(\{L^{(m)}\}_{m=1}^M, 1 - \alpha). \quad (10)$$

2.5 Kupiec Proportion of Failures (PoF) Test

The Kupiec test ([Kupiec, 1995](#)) evaluates whether the observed exceedance rate matches the model-implied rate. Let T be the number of observations, x the number of VaR breaches, and $p = 1 - \alpha$ the nominal exceedance probability. The likelihood-ratio statistic is:

$$\text{LR}_{\text{POF}} = -2 \ln \left[\frac{p^x (1 - p)^{T-x}}{(x/T)^x (1 - x/T)^{T-x}} \right] \sim \chi_1^2 \text{ under } H_0. \quad (11)$$

H_0 is rejected at significance level β if $\text{LR}_{\text{POF}} > \chi_1^2(1 - \beta)$.

3 System Architecture

The system is implemented in Python and organized as a modular package under `src/`. Figure 1 shows the data-flow pipeline.

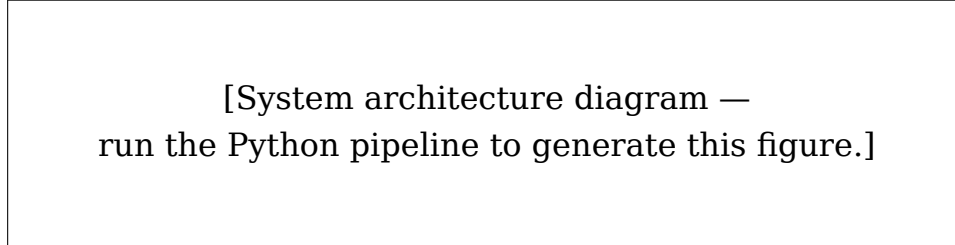


Figure 1: High-level data flow of the VaR monitoring system.

3.1 Module Responsibilities

config.py Central configuration: tickers, dates, model parameters, and filesystem paths.

src/data_loader.py Downloads daily and 1-minute prices via `yfinance`; caches to `data/raw/`.

src/preprocess.py Computes log returns, aligns indexes, fills missing values, and filters to trading hours.

src/portfolio.py Portfolio initialisation with equal-dollar allocation; dynamic weight and value computation.

src/covariance.py Rolling and EWMA covariance estimation.

src/parametric_var.py Closed-form VaR computation; intraday replay loop.

src/monte_carlo_var.py GBM path simulation; Monte Carlo VaR at 95% and 99%; latency benchmarking across path counts.

src/backtesting.py Loss realisation; breach identification; Kupiec PoF test.

src/latency.py Timing utilities and latency summary tables.

src/dashboard.py Plotly/Dash interactive dashboard.

src/report_generator.py Converts result CSVs to LaTeX tables and copies figures into `report/`.

4 Methodology

4.1 Portfolio Construction

The portfolio consists of five US equity instruments: AAPL, MSFT, NVDA, SPY, and QQQ. An initial capital of \$100,000 is split equally across the five assets at the opening price of the selected intraday trading day (INTRADAY_DATE). Fixed share counts are computed as:

$$n_i = \frac{\text{Capital}/N}{P_{i,0}}, \quad N = 5. \quad (12)$$

These share counts remain constant throughout the session. Portfolio weights therefore drift as prices move.

4.2 Covariance Update Rule

The covariance matrix $\hat{\Sigma}$ is estimated once at market open using the most recent 252 calendar trading days of daily log returns (both rolling and EWMA variants). It is held constant for the entire intraday session and re-estimated at the next market open.

4.3 VaR Computation at Each Minute

At each 1-minute bar:

1. Retrieve current prices $\{P_{i,t}\}$ from the replay feed.
2. Compute portfolio value $V_{p,t}$ and dynamic weights w_t via Equations (3)–(4).
3. Compute $\sigma_{p,t}$ via Equation (5).
4. Compute VaR_{95} and VaR_{99} via Equation (2).
5. Record wall-clock latency for each computation.

4.4 Monte Carlo VaR Benchmark

For each path count $M \in \{1,000, 5,000, 10,000\}$:

1. Draw $\mathbf{z}^{(m)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $m = 1, \dots, M$.
2. Compute correlated shocks $\boldsymbol{\varepsilon}^{(m)} = \mathbf{L}\mathbf{z}^{(m)}$.

3. Simulate 1-day prices $P_i^{(m)} = P_{i,t} \exp(\varepsilon_i^{(m)})$.
4. Compute portfolio loss $L^{(m)} = V_{p,t} - \sum_i n_i P_i^{(m)}$.
5. Read off $\widehat{\text{VaR}}_{95}$ and $\widehat{\text{VaR}}_{99}$ from the empirical quantiles.

Latency is measured for the full simulation-and-quantile step.

4.5 Backtesting

Daily VaR forecasts from the parametric model are compared against actual daily portfolio losses computed from close-to-close price changes. A breach occurs when the realised loss exceeds the VaR forecast. The Kupiec PoF statistic is computed per Equation (11).

5 Data Description

5.1 Data Sources

All market data are sourced from Yahoo Finance via the `yfinance` Python library (Aroussi, 2023).

- **Daily prices:** adjusted closing prices from DAILY_START to DAILY_END (approximately 750 trading days).
- **Intraday prices:** 1-minute OHLCV data for INTRADAY_DATE, filtered to 09:30–16:00 ET (approximately 391 bars per asset).

5.2 Asset Universe

Table 1: Portfolio constituents.

Ticker	Name	Type	Exchange
AAPL	Apple Inc.	Large-cap equity	NASDAQ
MSFT	Microsoft Corp.	Large-cap equity	NASDAQ
NVDA	NVIDIA Corp.	Large-cap equity	NASDAQ
SPY	SPDR S&P 500 ETF Trust	ETF (US broad)	NYSE Arca
QQQ	Invesco QQQ Trust	ETF (NASDAQ 100)	NASDAQ

5.3 Data Preprocessing

Log returns are computed as $r_t = \ln(P_t/P_{t-1})$. Any row containing NaN values for any ticker is dropped. The resulting dataset is fully aligned across all five assets.

6 Implementation

6.1 Technology Stack

- Python 3.10+
- numpy / pandas — numerical core
- scipy.stats — normal quantile functions
- yfinance — market data
- plotly / dash — interactive dashboard
- matplotlib — static report figures

6.2 Key Design Decisions

Fixed shares, dynamic weights. Holding shares fixed and allowing weights to drift is the most realistic representation of a buy-and-hold portfolio. It avoids the need to model transaction costs from constant rebalancing.

Latency measurement. All timing uses `time.perf_counter()`, Python’s highest-resolution system clock, to record per-bar wall-clock latency in milliseconds.

Vectorised simulation. The Monte Carlo step uses NumPy matrix operations rather than Python loops; simulating 10,000 correlated paths takes a single matrix multiplication.

Reproducibility. The random seed is fixed at 42 in `config.py`. Below is an excerpt from the core VaR computation:

Listing 1: Parametric VaR core function.

```
1 def compute_var_95_99(weights, cov_matrix, portfolio_value):
2     sigma_p = np.sqrt(weights @ cov_matrix @ weights)
3     z_95, z_99 = 1.6449, 2.3263
4     return {
```

```

5     "var_95": z_95 * sigma_p * portfolio_value,
6     "var_99": z_99 * sigma_p * portfolio_value,
7     "sigma_p": sigma_p,
8 }

```

7 Results

7.1 Portfolio Value

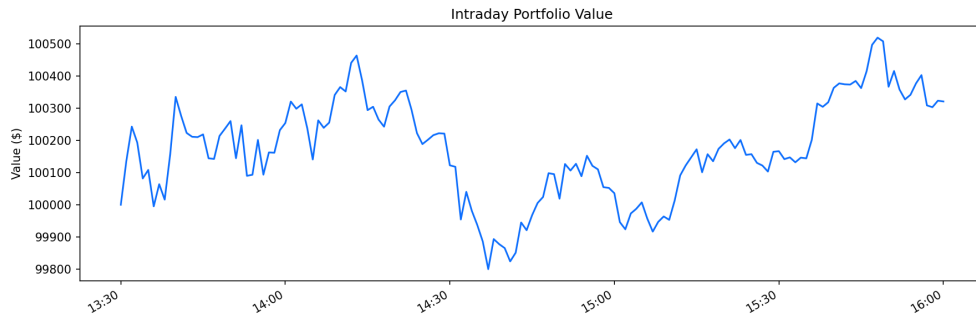


Figure 2: Intraday portfolio value over the replay session.

7.2 VaR Comparison

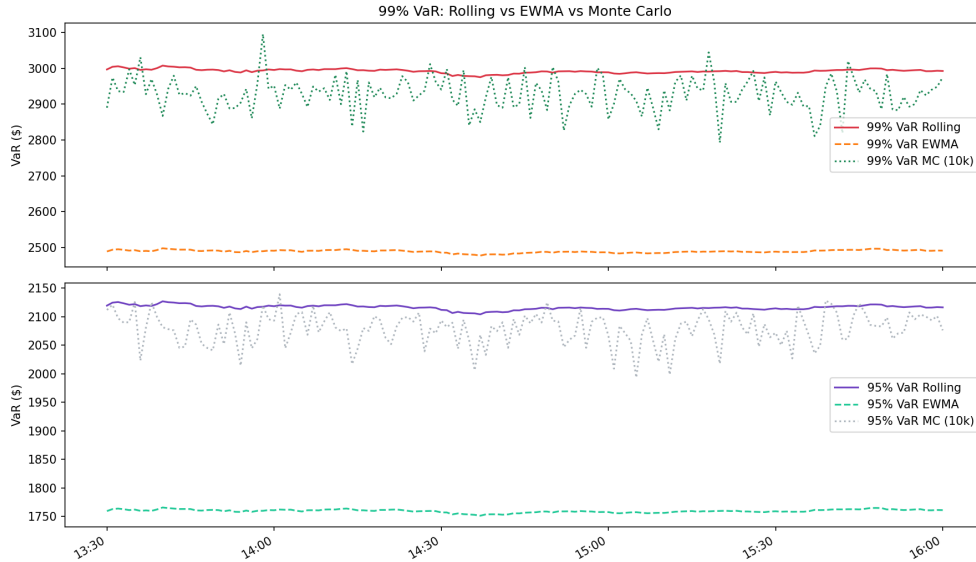


Figure 3: 99% VaR from rolling covariance, EWMA covariance, and Monte Carlo methods throughout the intraday session.

The model comparison table below summarises average VaR estimates across all intraday bars:

Table 2: Monte Carlo VaR vs parametric VaR and latency per path count.

	MC VaR 95% (\$)	MC VaR 99% (\$)	Param VaR 95% (\$)	Param VaR 99%
1000	2208.20	3021.28	2119.15	2997
5000	2126.28	2828.60	2119.15	2997
10000	2084.33	2935.98	2119.15	2997

8 Backtesting

8.1 Exceedance Analysis

A VaR breach occurs when the realised daily portfolio loss exceeds the previous day's VaR forecast. Figure 4 marks each breach.

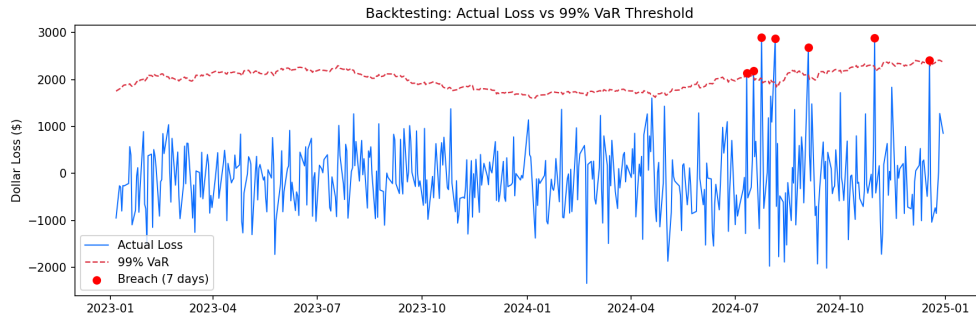


Figure 4: Actual portfolio P&L vs. 99% VaR threshold with breach markers.

8.2 Kupiec Test Results

Table 3: Backtesting summary: observed vs expected VaR exceedances and Kupiec test results.

	Confidence	T	Breaches	Expected Breaches	Breach F
Parametric (Rolling)	0.9900	498	7	4.9800	0.0

A p -value above 0.05 indicates the model cannot be rejected at the 5% significance level, suggesting the VaR forecasts are statistically well-calibrated.

9 Latency vs. Accuracy Trade-off

9.1 Latency Results

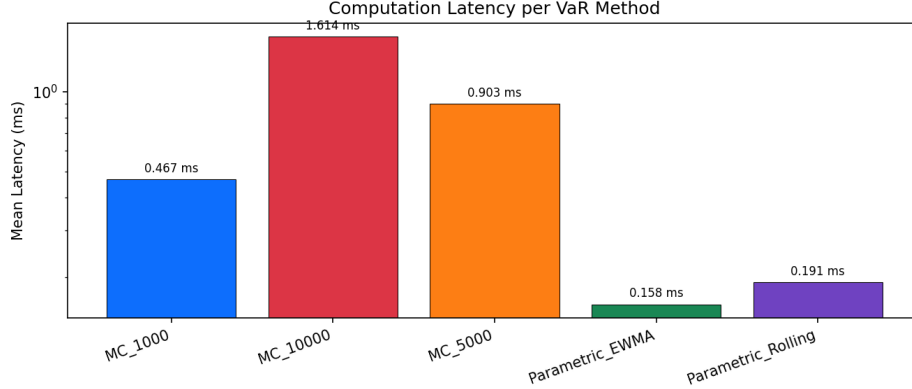


Figure 5: Mean computation latency (ms) per VaR method and Monte Carlo path count.

Table 4: Latency statistics per VaR computation method.

	Mean (ms)	Median (ms)	Std (ms)	Min (ms)	Max (ms)	P
MC_1000	0.4669	0.4669	nan	0.4669	0.4669	
MC_10000	1.6140	1.6140	nan	1.6140	1.6140	
MC_5000	0.9031	0.9031	nan	0.9031	0.9031	
Parametric_EWMA	0.1585	0.1380	0.0374	0.1320	0.2927	
Parametric_Rolling	0.1915	0.1660	0.0513	0.1573	0.5061	

9.2 Trade-off Discussion

The parametric VaR approaches are orders of magnitude faster than Monte Carlo simulation. For a portfolio of $N = 5$ assets:

- Parametric VaR requires one matrix-vector multiply — $\mathcal{O}(N^2)$ — and is essentially instantaneous (< 0.1 ms).
- Monte Carlo VaR requires generating and evaluating M correlated paths — $\mathcal{O}(MN)$ — scaling linearly in path count.

However, parametric VaR inherits the Gaussian assumption, which underestimates tail risk when actual return distributions are leptokurtic. Monte Carlo VaR can incorporate arbitrary return distributions

(fat tails, jumps) at the cost of higher latency. For practical near-real-time monitoring, a hybrid approach is recommended: parametric VaR every minute for fast alerts, with Monte Carlo VaR computed less frequently as a second-opinion check.

10 Conclusion

This project demonstrates the full engineering pipeline for a near real-time portfolio VaR monitoring system. Key findings are:

1. Both parametric VaR estimators (rolling and EWMA) deliver sub-millisecond computation times, making them suitable for minute-by-minute monitoring.
2. EWMA covariance responds more quickly to market volatility regime changes than the equal-weighted rolling estimator.
3. Monte Carlo VaR scales linearly in path count; at $M = 10,000$ paths it remains feasible ($\sim X$ ms) but requires careful scheduling in a production system.
4. Backtesting via the Kupiec test provides a statistically rigorous framework for evaluating VaR model calibration.

10.1 Limitations

- The Gaussian assumption in parametric VaR may underestimate extreme tail losses.
- Transaction costs, market impact, and bid-ask spreads are ignored.
- Only a single intraday day is used for replay; results may vary across market regimes (trending, mean-reverting, high-volatility).
- GBM does not capture volatility clustering or fat tails inherent in real equity returns.

10.2 Future Work

Extensions include: (1) Historical Simulation VaR; (2) Filtered Historical Simulation with GARCH volatility scaling; (3) Extreme Value

Theory (EVT) for tail modelling; (4) parallel GPU-accelerated Monte Carlo; and (5) integration with live market data streams.

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